Practical exercise 9

The following four data sets have been considered in lectures. Here you should use `R` to perform the relevant chi-squared tests.

1. Accidents to 414 workers over a 3 month period:

Number of accidents	0	1	2	3	4	5	≥ 6	Total
Frequency observed	296	74	26	8	4	6	0	414

   Conduct a goodness-of-fit test to determine whether a Poisson model can be used to describe the data.

   You can put the data into `R` in various ways – e.g. use

```
n.acc = c(rep(0,296), rep(1,74), rep(2,26), rep(3,8), rep(4,4), rep(5,6))
```

   Then obtain an estimate of the Poisson parameter (say `lambda.hat`) and compute the expected frequencies under the Poisson model. For example, you can do this in `R` using the following code:

```
pr.x = dpois(c(0:5), lambda.hat)
exp.f = pr.x*414
```

   Note that the frequency vector above does not include the frequency for ≥ 6, so we need to obtain this separately. Try the following commands and make sure you understand what they do:

```
pr.x = c( pr.x, (1-ppois(5,lambda.hat)) )
exp.f = c(exp.f, pr.x[7]*414)
```

   Then you will need to put the observed frequencies in a vector in `R` (say `obs.f`) , combine cells as necessary and compute the χ^2 statistic and p-value (check your answers with results given in lectures).

   Also note that the `chisq.test()` function in `R` can provide a gof test, but only in limited cases. First get some help on the use of the command (`?chisq.test`) and then try the following:

```
chisq.test(obs.f, p = pr.x)
```

   You should notice that this does not give the correct result in this case. There are two reasons for this:

   - it does not allow for adjusting/combining cells to ensure that we don't have small expected frequencies – to see this, try `chisq.test(obs.f, p=pr.x)$expected`

- it does not permit the number of estimated parameters to be accounted for in the degrees of freedom for the test (check df in output).

2. Eye colours of males visiting an optician, in four categories:

Colour	A	B	C	D
Frequency observed	89	66	60	85

Conduct a goodness-of-fit test to investigate whether the eye colour of a male visiting this optician belongs to category A, B, C or D with the same chance.

See lecture notes and follow a similar set of `R` commands as in Q1 above.

3. In a genetic plant breeding experiment a total of 1,500 plants were categorised into one of four classes (labelled C1, C2, C3 and C4) with the following results:

Class	C1	C2	C3	C4
Frequency	1071	62	68	299

We now wish to use a goodness-of-fit test to test the fit of the following model:

Class	C1	C2	C3	C4
Probability	$(2+\theta)/4$	$(1-\theta)/4$	$(1-\theta)/4$	$\theta/4$

where θ is an unknown parameter such that $0 < \theta < 1$.

Again, follow the method described in lectures and use `R` commands similar to those given in Q1.

For plotting the log-likelihood function, start with the entire range of θ, e.g.

```
theta = seq(0,1,length=50)
```

and then narrow this range down as necessary.

4. The data in file `cinema_data.txt` (on Canvas) give information on the gender of 75 students and whether or not they go to the cinema often.

Read the data into `R` as a data frame using the command `read.table(file, header=T)`

Then use the `table()` command to obtain the following 2-way classification:

	Not often	Often
Female	12	21
Male	20	22

Perform a χ^2 test to investigate whether the frequency with which students go to the cinema depends on their gender. Apply the test first without Yates' correction, and then with it.

Also, perform Fisher's exact test for the same hypothesis.